

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0644

Jim Bridger 138 kV
August 26, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0644. The Interconnection Customer proposes a 46.0 MW solar photovoltaic facility interconnecting at the Jim Bridger 138 kV substation in the Pennsylvania area. The request was received on July 15, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 202 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$52,915,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SOUTH HAMPTON to RAILYARD SUBSTATION 138kV	WALLENPAUPACK 500/345kV Transformer Bank	171.7	287	5.3
LAKIN MUNICIPAL to SOUTH HAMPTON 138kV	WALLENPAUPACK 500/345kV TX	158.1	316	5.3
SIMMS to GRAND SUBSTATION 138kV	LAKIN MUNICIPAL 345/138kV Transformer Bank	124.6	193	25.3
N DAYTON to GRAND 138kV Line	LAKIN MUNICIPAL 345 kV/138 kV XFMR	120.2	218	25.3
BURNS SUBSTATION to CORONADO 345kV	CLEAR SPRINGS 500/345 kV Transformer Bank	118.9	1,145	8.6
PRICES FORK SUBSTATION to CLEAR SPRINGS 138kV	WARM SPRINGS to LAKIN MUNICIPAL 345kV Line	118.4	171	5.0
JIM BRIDGER to NORTH DAYTON 138kV L	LAKIN MUNICIPAL 345/138 kV Transformer Bank	117.4	234	25.3
INGHAMS to BOEING 138kV	WARM SPRINGS to KOOSKIA 138kV	110.4	240	5.5
BURNS to HILLS CHAPEL 345kV	CLEAR SPRINGS 500/345kV TX	110.0	1,468	9.7
WHEELABRATOR FRACKVILLE ENERGY 500 kV/1 kV Transformer	TALENENERGY MARTINS CREEK LLC HARWOOD to KLINE TOWNSHIP COGEN FACILITY 345kV L	108.6	899	5.1
WALLENPAUPACK 500-345 kV TX	TOMBSTONE to BIXBY HILL 500kV	108.0	1,606	7.9
HOHOKAM SUBSTATION to PRICES FORK 138kV line	LAKIN MUNICIPAL to WARM SPRINGS 345kV	107.0	202	5.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WNP NO. 2 CONSTRUCTION to LAKIN MUNICIPAL 138kV	LAKIN MUNICIPAL 345/138kV Autotransformer	106.8	265	18.6
WNP NO. 2 CONSTRUCTION to SIMMS 138kV	LAKIN MUNICIPAL 345/138kV XFMR	106.3	215	25.3
HOHOKAM SUBSTATION to SAN FRANCISQUITO 1 138kV	LAKIN MUNICIPAL to WARM SPRINGS 345kV	105.7	215	5.0
WARM SPRINGS to TALENENERGY MARTINS CREEK LLC HARWOOD 345kV	TALENENERGY MARTINS CREEK LLC HARWOOD 345-1 kV TX	100.5	1,070	16.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SOUTH HAMPTON to RAILYARD SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$3,928,000	100% (\$3,928,000)
LAKIN MUNICIPAL to SOUTH HAMPTON 138kV	Rebuild line with higher-capacity conductor	\$23,465,000	100% (\$23,465,000)
GRAND to SIMMS 138kV	Rebuild line with higher-capacity conductor	\$1,927,000	100% (\$1,927,000)
GRAND to N DAYTON 138kV line	Rebuild line with higher-capacity conductor	\$1,664,000	100% (\$1,664,000)
BURNS SUBSTATION to CORONADO 345kV	Rebuild line with higher-capacity conductor	\$21,335,000	1% (\$258,000)
PRICES FORK to CLEAR SPRINGS 138kV line	Rebuild line with higher-capacity conductor	\$3,736,000	100% (\$3,736,000)
JIM BRIDGER to NORTH DAYTON 138kV L	Rebuild line with higher-capacity conductor	\$2,566,000	100% (\$2,566,000)
BOEING to INGHAMS 138kV	Rebuild line with higher-capacity conductor	\$10,350,000	4% (\$412,000)
BURNS to HILLS CHAPEL 345kV	Rebuild line with higher-capacity conductor	\$58,636,000	1% (\$756,000)
WHEELABRATOR FRACKVILLE ENERGY 500/1kV XFMR	Replace transformer with higher-rated unit	\$24,277,000	2% (\$540,000)
WALLENPAUPACK 500-345 kV TX	Replace transformer with higher-rated unit	\$38,979,000	1% (\$423,000)
HOHOKAM to PRICES FORK 138kV	Reconductor line	\$1,445,000	100% (\$1,445,000)
LAKIN MUNICIPAL to WNP NO. 2 CONSTRUCTION 138kV	Reconductor line	\$1,514,000	31% (\$463,000)
SIMMS to WNP NO. 2 CONSTRUCTION 138kV L	Reconductor line	\$1,979,000	100% (\$1,979,000)
HOHOKAM SUBSTATION to SAN FRANCISQUITO 1 138kV	Reconductor line	\$8,274,000	100% (\$8,274,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TALENENERGY MARTINS CREEK LLC HARWOOD to WARM SPRINGS 345kV Line	Reconductor line	\$17,982,000	6% (\$1,079,000)
	Total allocated to project		\$52,915,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.